GreenFlow: A Vision-Based Smart Irrigation System Using Crop Color Analysis

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ABSTRACT Watering a whole field in the same way wastes water, because the parts of a field rarely dry out at the same speed. This paper introduces GreenFlow, a Python prototype that decides where to irrigate from the color of grass and crops. A camera image of the field is divided into zones, the color of each zone is converted to the HSV space and scored with the excess-green index. Zones that stay green are marked healthy and receive little or no water, while zones that turn yellow or brown are marked as stressed and are watered first. The decision is sent to a microcontroller that opens only the valves of the stressed zones, and a web dashboard shows the zone map and the water log. The paper gives the motivation, a short review of related work, and the design of the system. The review shows that soil sensing, plant-color analysis, and valve control are usually studied separately. GreenFlow is a prototype for decision support. Its thresholds are rule-based, and the water saving compared with a fixed timer has not yet been measured in a real field. The accuracy of the color score under changing sunlight and on real crops still has to be validated.

INDEX TERMS smart irrigation, precision agriculture, image processing, HSV color analysis, vegetation index, water conservation.

I. INTRODUCTION

Every crop needs water, but not every part of a field needs it at the same time. Agriculture is the largest user of freshwater in the world, and in countries such as Pakistan, where water is limited, wasted irrigation is both a cost and a risk. If one area is watered too much, water is lost, and if another area is watered too little, the plants there are stressed and the yield falls.

Most small farms and lawns still water by hand or with a fixed timer. A timer does not know which area is already green and which area is drying out. Soil sensors can measure this, but each one reads only the point where it is placed, so a whole field needs many of them. A camera, on the other hand, sees the whole field at once. Healthy leaves look green, and water stress often turns them yellow or brown. A camera cannot say why a color has changed, but it can show where to look first. So we decided to turn the color of the plants directly into a watering decision.

For the camera part, each image is converted to the HSV color space [1], which separates color from brightness, and a green score is computed for each pixel with the excess-green index [2]. The image is split into a grid of zones, and the mean score of a zone decides its state. For the control part, a simple set of rules maps each state to a watering time, and a microcontroller switches the valve of that zone.

This paper presents GreenFlow, which connects these ideas in one system. Its main contributions are: (i) a color analysis module that scores every zone of the field, (ii) a rule-based controller that waters only the zones that need it, (iii) a tool for comparing the proposed watering with a fixed timer, and (iv) a dashboard that clearly separates observed, simulated, and demo data. GreenFlow is a prototype for decision support and is not a certified agricultural system. Section II reviews related work.

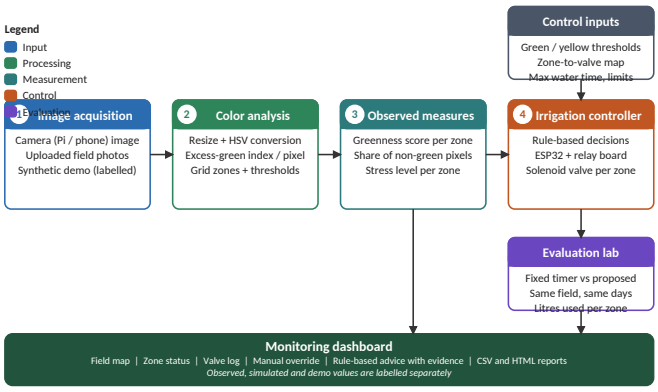


Fig. 1. Architecture of GreenFlow: field images feed the color analysis module, which drives the irrigation controller, and both report to one monitoring dashboard.

II. LITERATURE REVIEW

A. Measuring Water Need in the Field

Water need has traditionally been judged by hand, for example by feeling the soil, or estimated from a fixed schedule. Sensor-based systems improved on this. Kim et al. built a distributed wireless sensor network for a center-pivot system, so that sensing stations in the field could guide where water was applied [3]. Gutierrez et al. used soil-moisture and temperature sensors with a wireless gateway to open valves automatically when the soil dropped below a threshold [4]. Their work showed that automatic control is practical. However, each sensor reads only one point, so covering a large field needs many sensors, and the readings do not show how the whole area looks.

B. Plant Color and Vegetation Indices

Healthy leaves look green because chlorophyll absorbs red and blue light and reflects green. Woebbecke et al. compared simple color indices, including excess green, to separate plants from soil and residue under changing light [2]. Gitelson et al. proposed an index that uses the green channel to measure vegetation with ordinary visible-light cameras [5]. The HSV space is often used for such work, because the hue stays similar when the brightness changes [1]. These methods are cheap and cover a whole area, but color is only a sign of stress and not a proof of it. Jones reviewed plant-based irrigation scheduling and noted that plant signs can also be affected by factors other than water [6]. For this reason color should guide where to water, and limits on the watering time are still needed.

C. Automated Irrigation Control

Automated systems usually pair a microcontroller with relays and solenoid valves, and they decide by simple threshold rules [3], [4]. As with any rule-based system, the result depends on the thresholds chosen, so they are most useful when they are set from real observation of the crop and the soil.

D. Research Gap

Table I summarises the approaches. Soil sensing, plant-color analysis, and valve control each solve part of the problem. In the studies we reviewed, they are reported separately, and we did not find a small open tool that carries one field image from color measurement to a zone-level valve decision and a water log. Our review is short and not exhaustive, but this gap motivated GreenFlow.

Table I. Summary of related approaches

Approach	Strength	Limitation
Fixed timer	Simple and cheap	Ignores plant condition
Soil-sensor networks [3], [4]	Direct moisture reading	Point readings; many sensors needed
Color indices [2], [5]	Non-contact; covers whole area	Light sensitive; color is only a sign
GreenFlow	Links image color to zone valves	Not yet validated in a real field

REFERENCES

[1] A. R. Smith, “Color gamut transform pairs,” *ACM SIGGRAPH Comput. Graph.*, vol. 12, no. 3, pp. 12–19, 1978.

[2] D. M. Woebbecke, G. E. Meyer, K. Von Bargen, and D. A. Mortensen, “Color indices for weed identification under various soil, residue, and lighting conditions,” *Trans. ASAE*, vol. 38, no. 1, pp. 259–269, 1995.

[3] Y. Kim, R. G. Evans, and W. M. Iversen, “Remote sensing and control of an irrigation system using a distributed wireless sensor network,” *IEEE Trans. Instrum. Meas.*, vol. 57, no. 7, pp. 1379–1387, 2008.

[4] J. Gutierrez, J. F. Villa-Medina, A. Nieto-Garibay, and M. A. Porta-Gandara, “Automated irrigation system using a wireless sensor network and GPRS module,” *IEEE Trans. Instrum. Meas.*, vol. 63, no. 1, pp. 166–176, 2014.

[5] A. A. Gitelson, Y. J. Kaufman, and M. N. Merzlyak, “Use of a green channel in remote sensing of global vegetation from EOS-MODIS,” *Remote Sens. Environ.*, vol. 58, no. 3, pp. 289–298, 1996.

[6] H. G. Jones, “Irrigation scheduling: Advantages and pitfalls of plant-based methods,” *J. Exp. Bot.*, vol. 55, no. 407, pp. 2427–2436, 2004.